

Matplotlib Basics

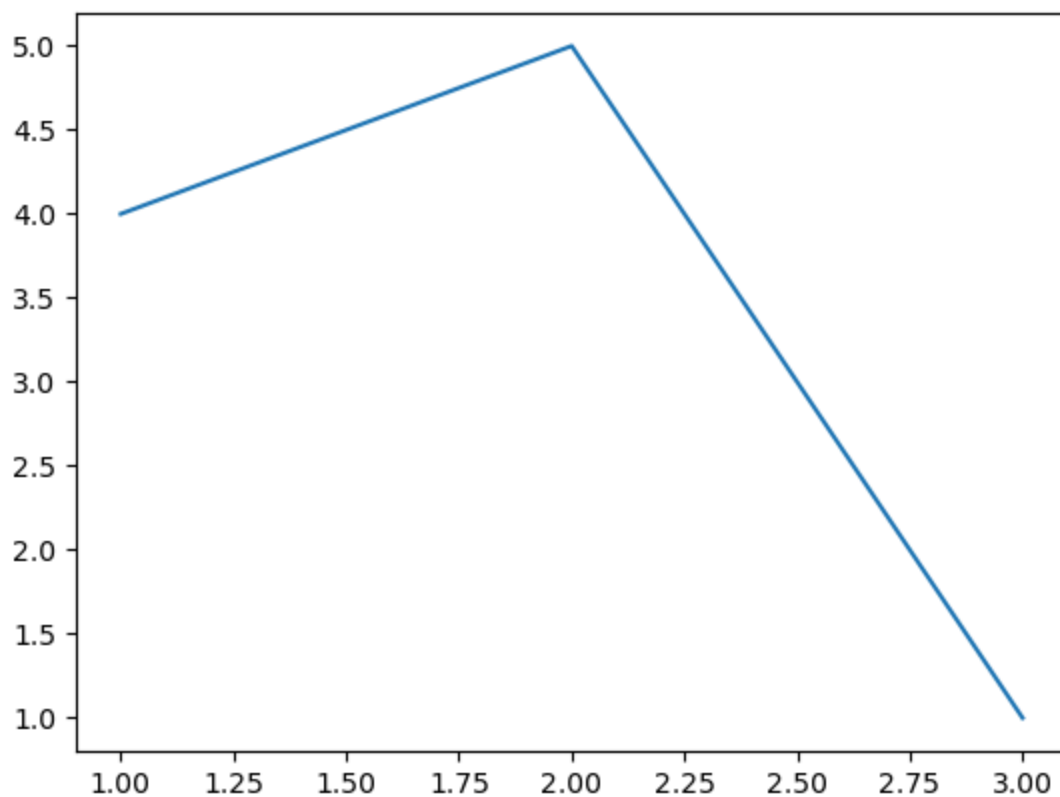
First Visualisation

```
In [1]: #importing pyplot

#from matplotlib import pyplot as plt
import matplotlib.pyplot as plt

#plotting to our canvas
plt.plot([1,2,3],[4,5,1])

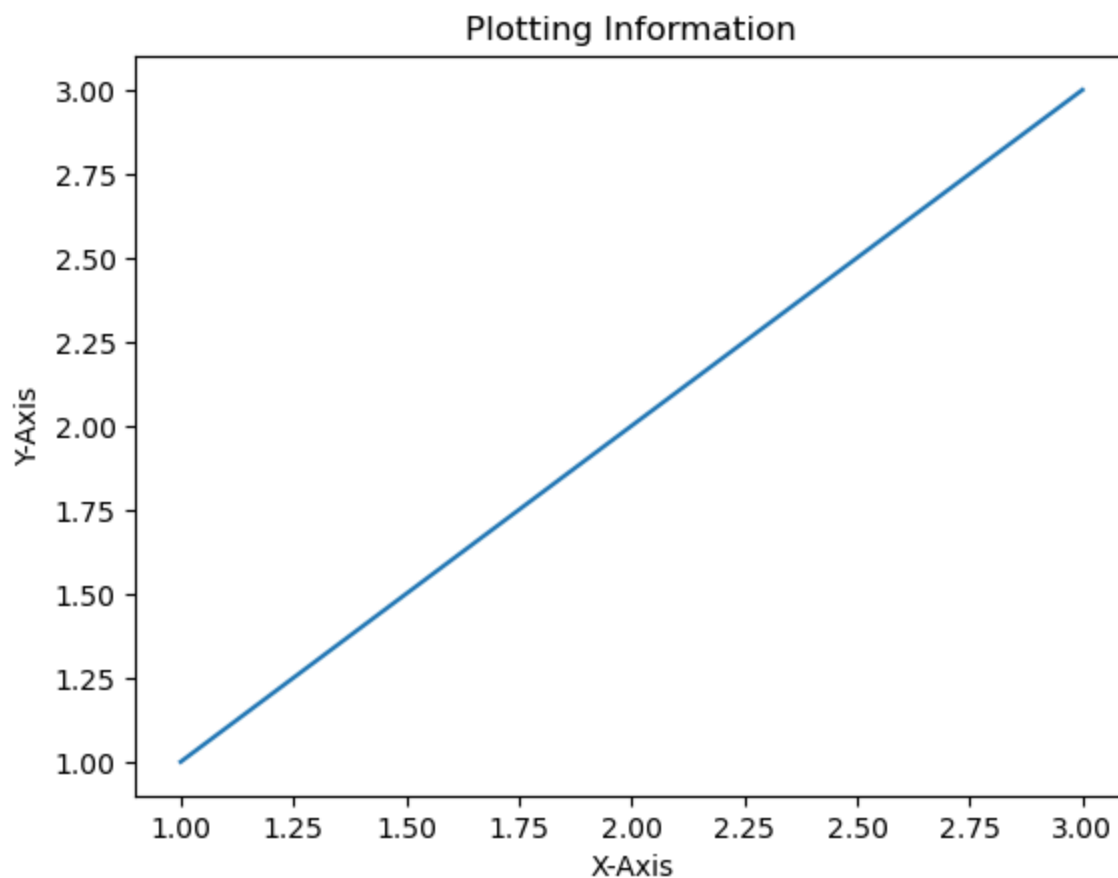
#Showing What we Plotted
plt.show()
```



```
In [4]: #We Can Also Pass Through Lists
X = [1,2,3]
Y = [1,2,3]
plt.plot(X,Y)

#Putting Title
plt.title('Plotting Information')
#Putting Label to the Axis
plt.xlabel('X-Axis')
plt.ylabel('Y-Axis')

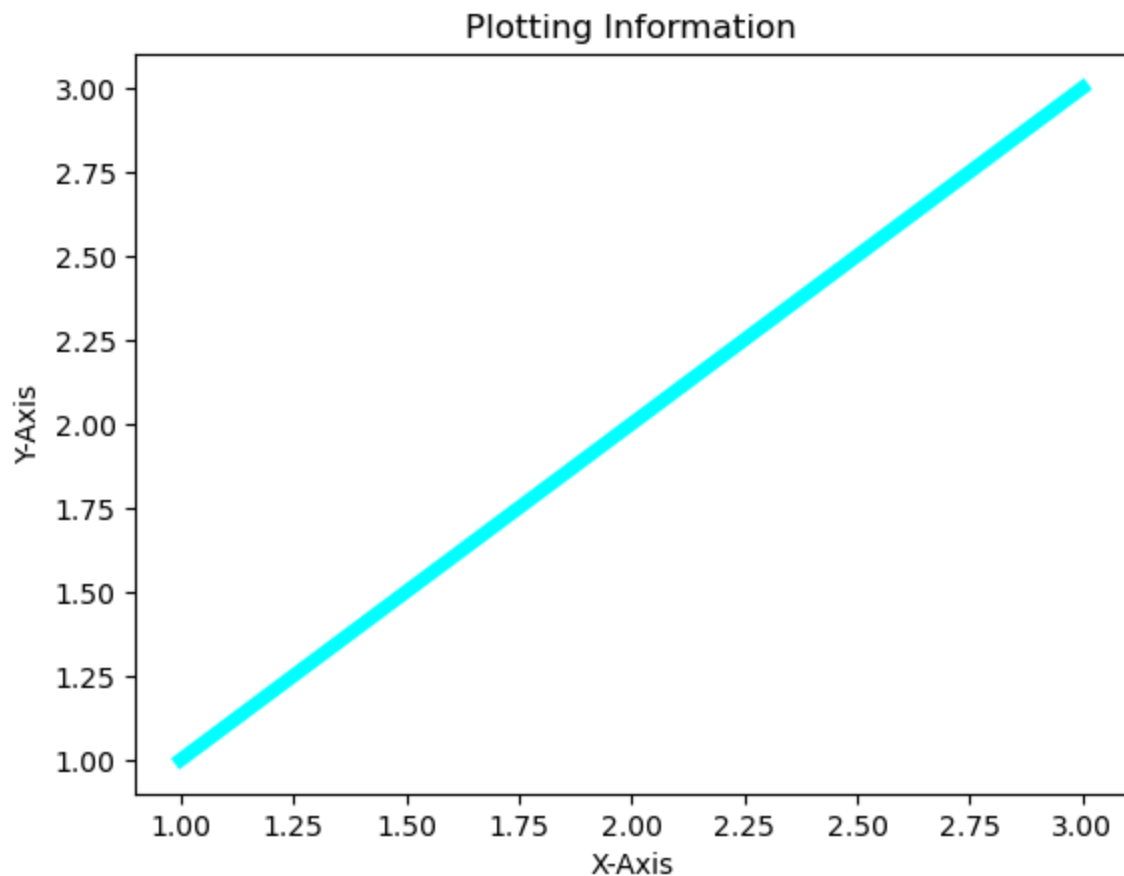
plt.show()
```



```
In [6]: #We Can Also Pass Through Lists
X = [1,2,3]
Y = [1,2,3]
plt.plot(X,Y, color='cyan', linewidth=5)

#Putting Title
plt.title('Plotting Information')
#Putting Label to the Axis
plt.xlabel('X-Axis')
plt.ylabel('Y-Axis')

plt.show()
```

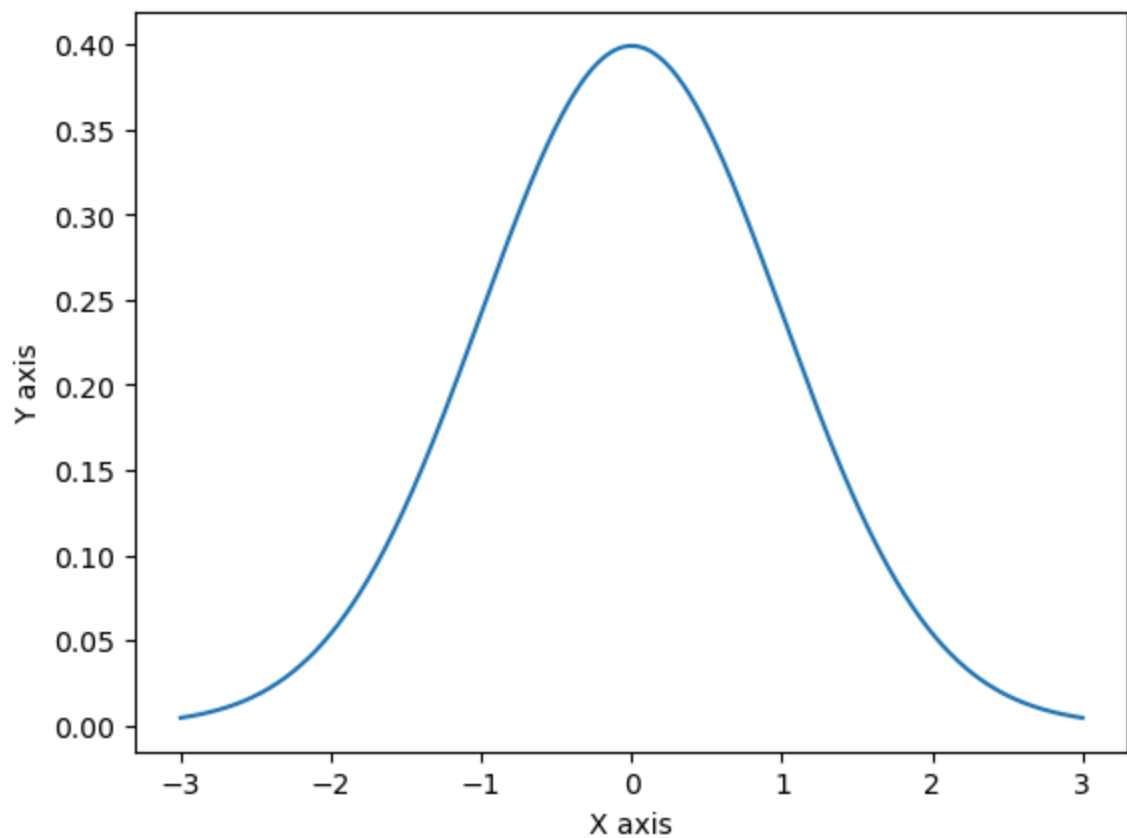


Draw A Line Graph

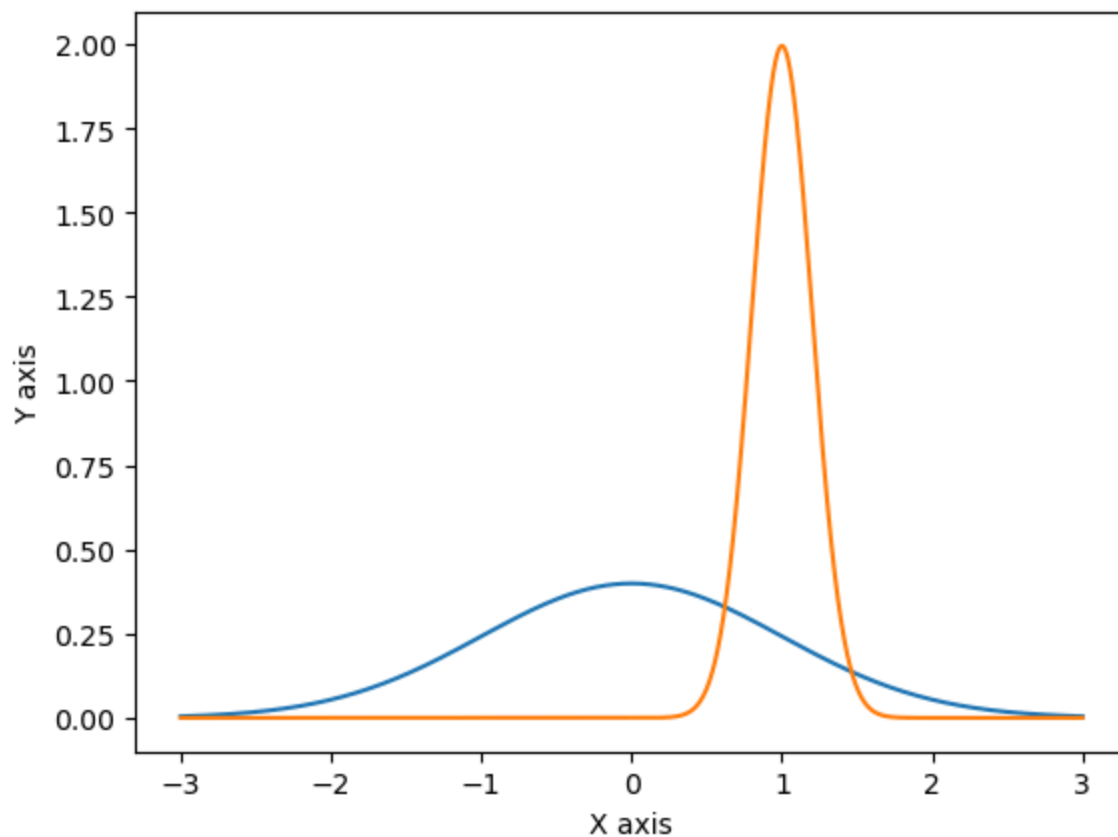
```
In [7]: from scipy.stats import norm #For normal distribution
import matplotlib.pyplot as plt
import numpy as np

x = np.arange(-3, 3, 0.001) # np.arange(start, stop, step) creates a NumPy array

plt.plot(x, norm.pdf(x)) # evaluates the probability density function (PDF) of a
plt.ylabel('Y axis')
plt.xlabel('X axis')
plt.show()
```



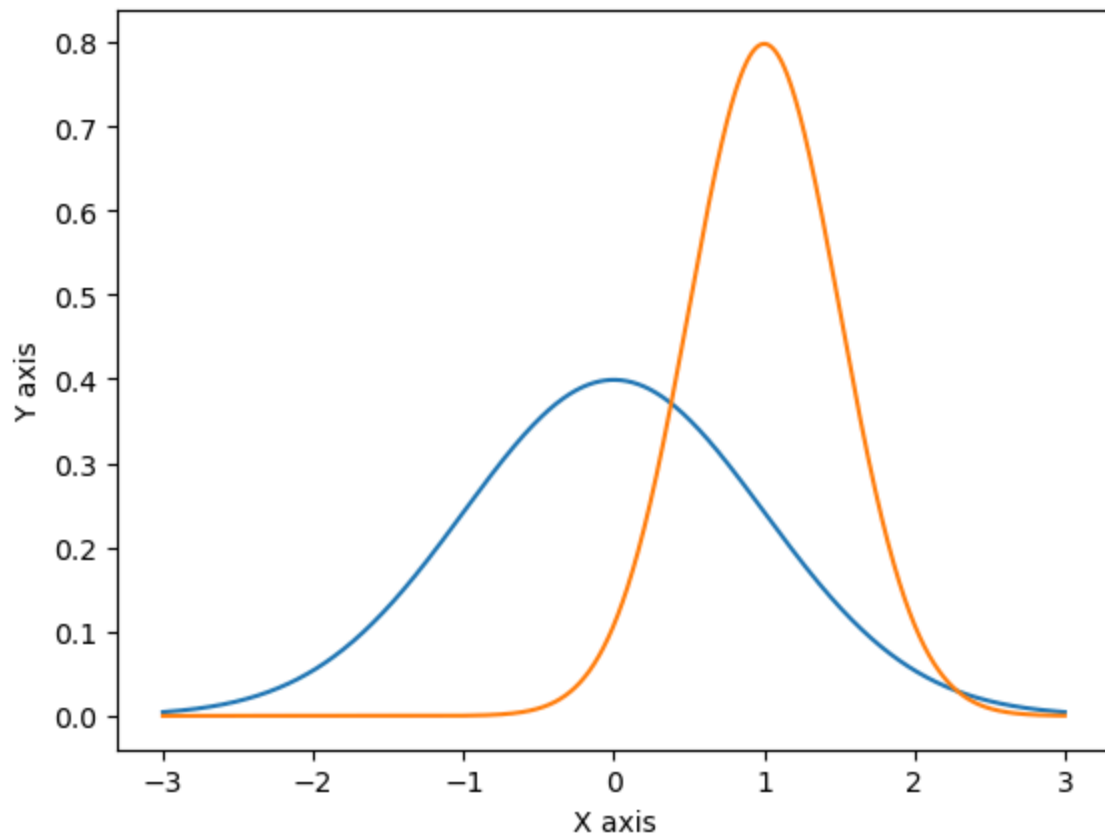
```
In [8]: plt.plot(x, norm.pdf(x))  
#print(norm.pdf(x))  
  
#print (norm.pdf(x, 1.0, 0.5))  
plt.plot(x, norm.pdf(x, 1.0, 0.2)) #mean=1.0 and Std deviation=0.2  
  
plt.ylabel('Y axis')  
plt.xlabel('X axis')  
  
plt.show()
```



Save it to A File

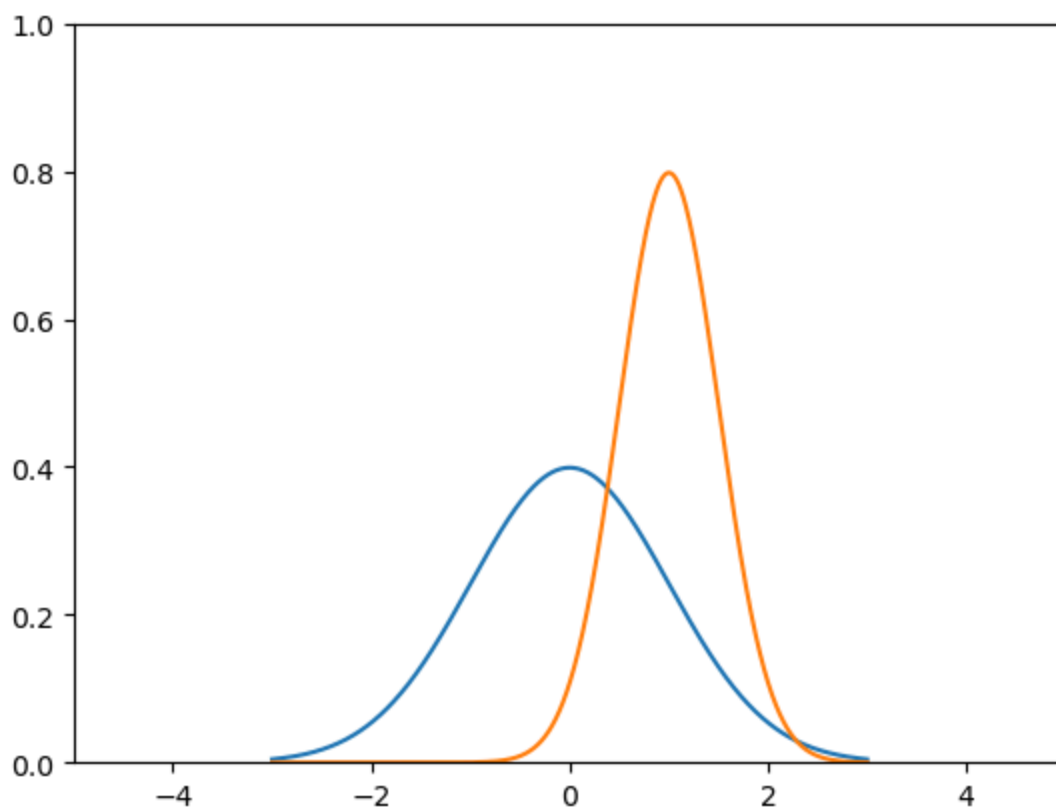
```
In [10]: plt.plot(x, norm.pdf(x))  
plt.plot(x, norm.pdf(x, 1.0, 0.5))  
plt.savefig(r'D:\24BCSH03\MyPlot1.png', format='png')  
  
plt.ylabel('Y axis')  
plt.xlabel('X axis')
```

Out[10]: Text(0.5, 23.52222222222222, 'X axis')

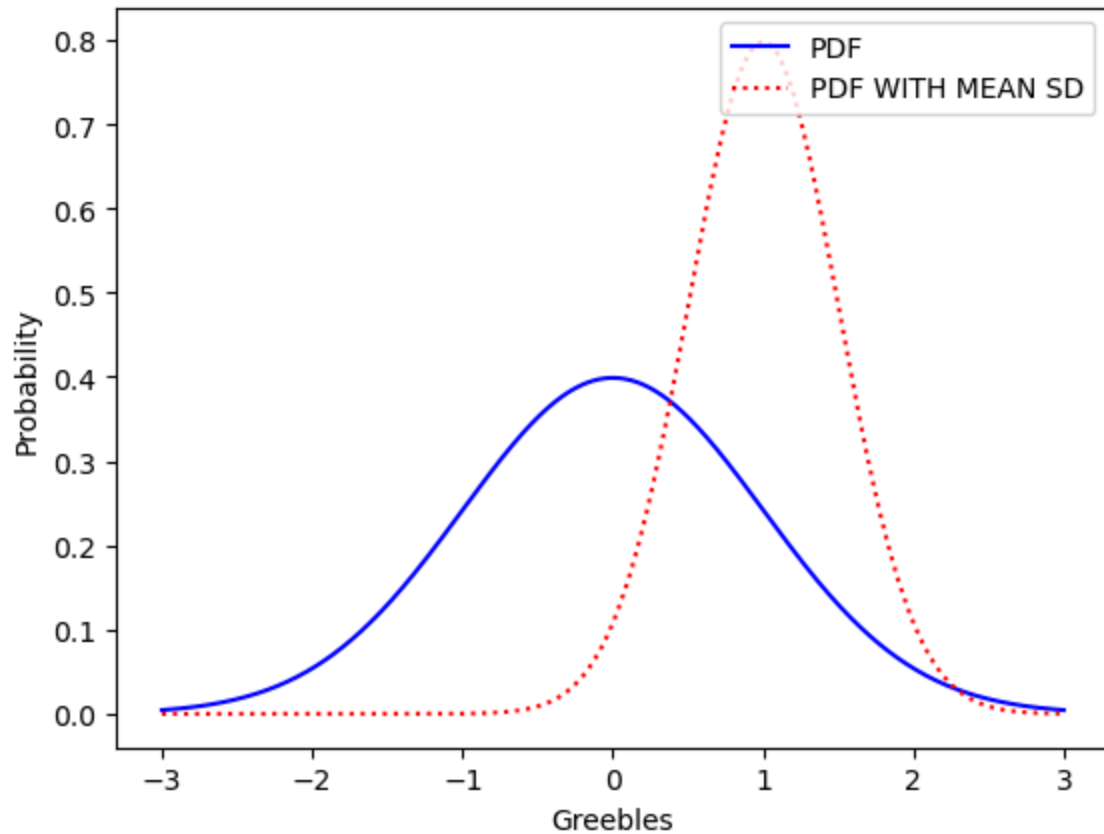


Adjusting the Axes

```
In [11]: axes = plt. axes ()  
axes.set_xlim([-5, 5])  
axes.set_ylim([0, 1.0])  
#axes. set_xticks([-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5])  
#axes.set_yticks([0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0])  
plt.plot(x, norm.pdf(x))  
plt.plot(x, norm.pdf(x, 1.0, 0.5)) #mean=1.0 and standard deviation=0.5  
plt.show()
```

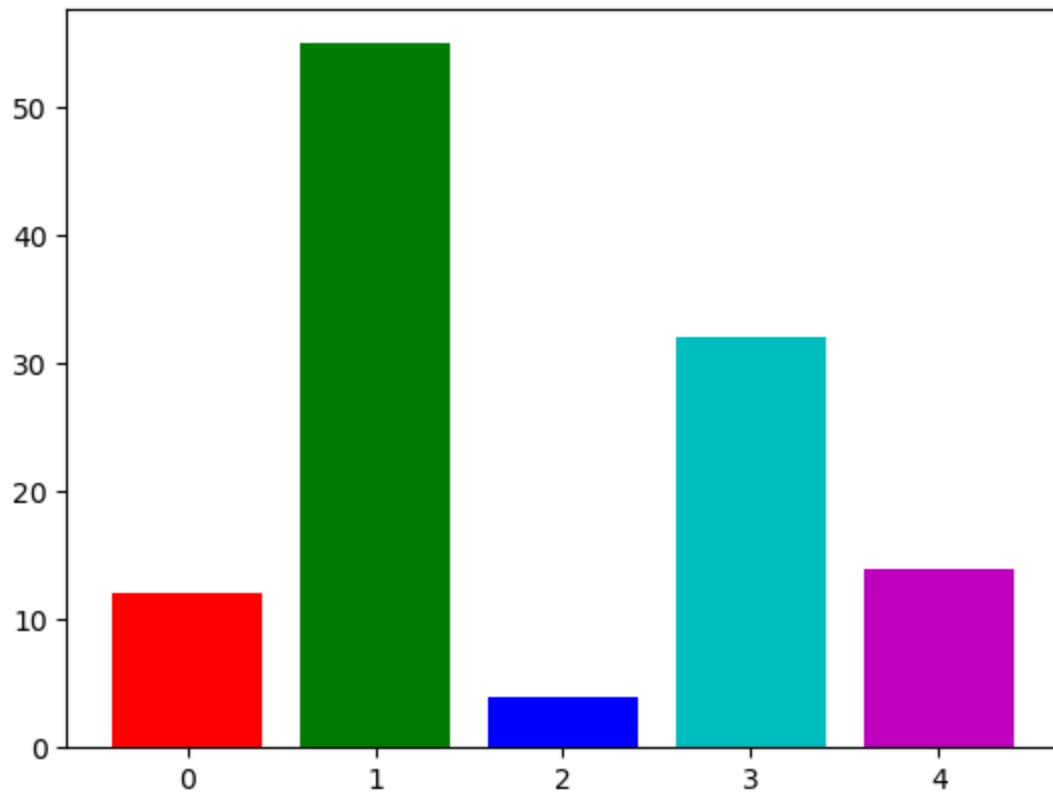


```
In [15]: axes.set_xticks([-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5])
axes.set_yticks([0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0])
axes.grid()
plt.xlabel('Greebles')
plt.ylabel('Probability')
plt.plot(x, norm.pdf(x), 'b-')
plt.plot(x, norm.pdf(x, 1.0, 0.5), 'r:')
plt.legend(['PDF', 'PDF WITH MEAN SD'], loc=1) # For Middle of the Graph : Loc
plt.show()
```



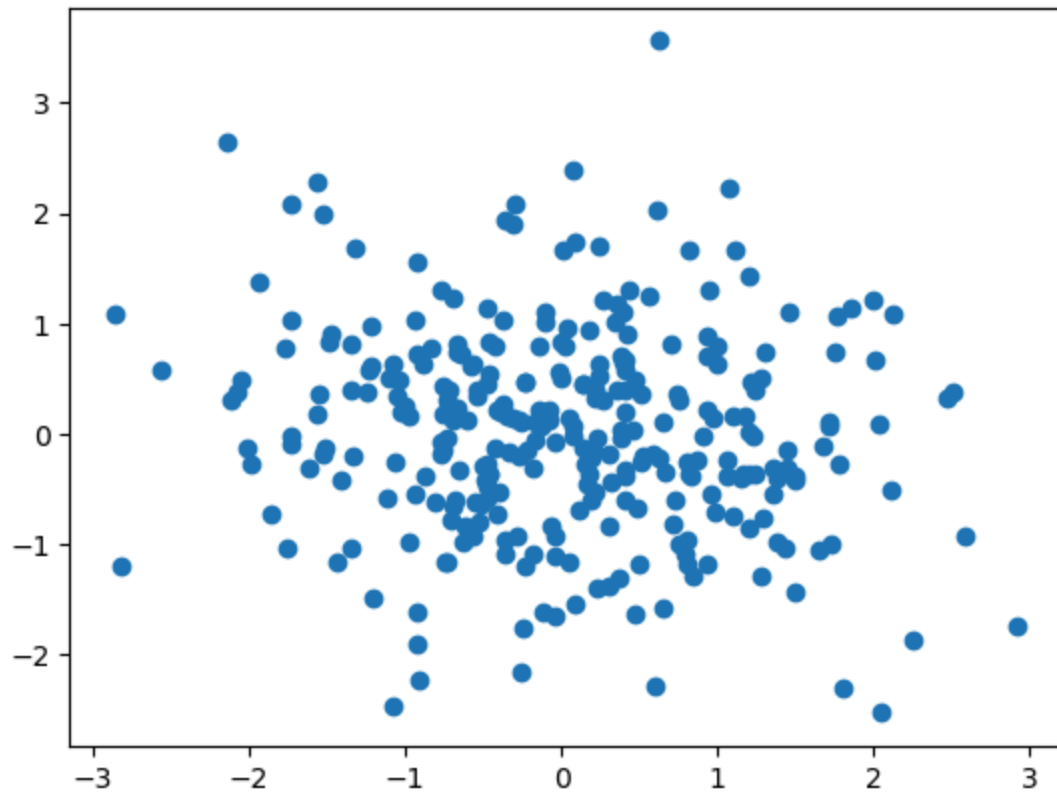
Bar Chart


```
In [16]: values = [12, 55, 4, 32, 14]
         colors = ['r', 'g', 'b', 'c', 'm']
         plt.bar(range(0,5), values, color= colors)
         plt.show()
```



Scatter Plotting

```
In [23]: from pylab import randn  
X = randn(300)  
Y = randn(300)  
plt.scatter(X,Y)  
plt.show()
```



In []: